# Rakshit Ratan Pathak

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## SUMMARY

Final-year B.Tech student (CGPA 8.3) who builds compiler and GPU tooling in C++: a C++20 kernel autotuner with MLIR passes (in progress), a 1,200+ line C++17 compiler with an optimizer and bytecode VM, and CUDA kernels analysed down to PTX. Looking for compiler and system-software internships, January–June 2027.

## EDUCATION

**Vellore Institute of Technology, Vellore** | B.Tech, Computer Science and Business Systems | **CGPA 8.3** 2023–2027

Class XII (CBSE), Delhi Public School Ranchi: 87% | Class X (CBSE), DAV Jharsuguda: 99.8%

**Relevant coursework:** Computer Organization and Architecture, Operating Systems, Formal Languages and Automata, Data Structures and Algorithms, Design and Analysis of Algorithms, Object-Oriented Programming, DBMS, Computer Networks.

## TECHNICAL SKILLS

- **Languages:** C++ (C++17/20), C, CUDA, Python, Java, TypeScript, SQL
- **Compilers and systems:** lexing, recursive-descent parsing, ASTs, constant folding, bytecode generation with backpatching, stack VMs, MLIR pass API and TableGen, CUDA Driver API, roofline and occupancy analysis, shared-memory bank conflicts, reading PTX
- **Tools:** gcc, clang, CMake, Make, Git, AddressSanitizer, UBSan, ThreadSanitizer, PyTorch, TensorFlow, PostgreSQL, Google Cloud

## PROJECTS

**PTXForge: GPU kernel autotuner and MLIR passes in C++20** (in progress) github.com/RakshitRPathak/ptxforge

- Built a C++20 autotuner for a parametric GEMM + bias + GELU CUDA kernel: 3,610-config tile search space, a roofline cost model (arithmetic intensity, register caps, occupancy, wave quantization) and closed-loop search. 16 unit tests, clean under ASan and UBSan.
- Modeled shared-memory bank conflicts (32 banks, vector-access phases) in one header used by both the tuner and an MLIR padding pass. It shows the kernel's transposed tile store is 8-way conflicted unpadded and conflict-free with 4 floats of padding.
- Wrote MLIR passes in C++ with TableGen (epilogue fusion on `linalg`, workgroup-memory padding) and a CUDA Driver API runner that loads PTX in-process and checks results against an fp64 reference before timing. LLVM and GPU validation in progress.
- Built an ablation harness comparing knowledge-graph, rule-based, random and cost-model-guided search at equal trial budgets over 10 seeds.

**minic: a compiler in C++17** github.com/RakshitRPathak/minic

- Built the full pipeline in 1,200+ lines: lexer, recursive-descent parser, AST, semantic analysis, optimizer, bytecode generator and stack VM. 43 tests (12 check exact error messages), clean under ASan and UBSan.
- Implemented constant folding and dead-branch and dead-loop elimination. Folding shares its arithmetic code with the VM, and optimized and unoptimized builds give identical output on all 14 test programs.
- Resolved forward jumps by backpatching (the same problem linkers solve with relocations), and ran semantic analysis before optimization so removing dead code cannot change which programs compile.

**cuda-gpu-bench: CUDA kernels and analysis** github.com/RakshitRPathak/cuda-gpu-bench

- Wrote naive and shared-memory tiled matrix multiply and three sum reductions (contended atomics, shared-memory tree, warp shuffle), with a harness that times with CUDA events and verifies against a CPU reference. Compiled them to PTX and counted loads, barriers and shuffles per kernel.
- Built a CPU emulator of threads, blocks, shared memory, barriers, atomics and warp shuffles to unit-test kernel logic without a GPU: 33 checks, and ThreadSanitizer flags a removed `__syncthreads()` as a data race.

Also on GitHub: agentic-bug-repair (a Java agent that localizes bugs on a code graph and verifies fixes in a sandbox, 147 tests), mini-kv-store (a crash-safe LSM-tree store, 39 tests) and a concurrency-safe PostgreSQL booking backend.

## EXPERIENCE

**Software Engineering Intern, aiDataWorks** May–Jul 2026

- Built and tested the LLM component of a healthcare data project (RAG, OpenAI APIs, prompt engineering) on enterprise Google Cloud.
- Worked on data-governance and security policies for three client engagements and took part in client interactions.

**Computer Science Intern, THDC India Limited** (Khurja Super Thermal Power Project) May–Jun 2025

- Gained exposure to SCADA/PLC systems, SQL/ERP, networking and cybersecurity.

## PATENTS AND LEADERSHIP

- Six patent applications filed through VIT's IPR Cell, five published (mine tremor detection, wildfire risk prediction, information-systems trust-decay auditing, opinion-lineage sentiment correction, risk-constrained emergency control).
- Secretary, IEEE Power & Energy Society student chapter, VIT.
- Karate black belt and assistant instructor.